

FedQ-Fraud: A Quantum-Enforced Federated Learning Framework for Financial Fraud Detection

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Abstract

The rapid growth of online financial transactions has made fraud detection a critical priority, especially with evolving fraud strategies that evade traditional systems. This paper presents a novel and secure fraud detection framework integrating Deep Learning (DL), Federated Learning (FL), and Quantum Encryption Communication (QEC). Our approach ensures high fraud detection accuracy while maintaining user data privacy and secure communication. We implemented a 3-layer GRU model using the MOON algorithm under the FL paradigm and achieved a global accuracy of 97.47%, outperforming traditional models like LSTM, 1D-CNN and XGBoost. To secure model parameter exchange between clients and server, we evaluated two entanglement-based Quantum Key Distribution (QKD) protocols — BBM92 and MDI-QKD. Experimental results revealed BBM92 to be more stable and suitable for integration with FL, demonstrating superior average secret key rate (SKR) and lower quantum bit error rate (QBER). The proposed system effectively combines accuracy, privacy, and quantum security, making it a scalable solution for real-world fraud detection.

Keywords

Fraud Detection, Federated Learning, Deep Learning, MOON Algorithm, BBM92 Protocol, Quantum Key Distribution, Financial Cybersecurity.

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1 Introduction

The rapid modernization of the financial sector has revolutionized online transactions, making them faster and more accessible with just a few clicks. However, this growth has also led to a significant rise in transaction fraud. As highlighted in the Nasdaq Verafin 2024 Global Financial Crime Report, global fraud losses amounted to an alarming \$485.6 billion. This underscores the urgent need to combat evolving fraudulent activities with advanced technological solutions. One such breakthrough is the advent of Machine Learning (ML), which enables systems to emulate human learning and adapt to dynamic environments. Broadly, ML algorithms operate through three key components: the decision process, the error function, and the model optimization process.

Sumedh et al. [4] performed a comparative analysis of various ML model architectures. They achieved 99.21% accuracy and 95.20% recall by using Support Vector Machine (SVM) to detect frauds in Credit Card Transactions. As fraud schemes grew more intricate, ML models could not cope with the increase, thus DL concepts started to rise. They are more complex because they are built like the human brain. Compared to ML, DL has a deep neural network structure to detect more intricate patterns, thus increasing its efficiency. There

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are mainly six types of DL models, they are: Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Autoencoders and Variational Autoencoders (AEs and VAEs), Generative Adversarial Networks (GANs), Diffusion Models and Transformer Models. Anish et al. [3] proposed a comparison of LSTM, CNN, and Hybrid LSTM-CNN to detect fraudulent bank transactions. They achieved an accuracy of 98.1% and a recall of 81.3% using Hybrid LSTM-CNN.

As things started to get complex, the models started becoming data-hungry. Thus, data privacy became a major concern as previously data was transferred to servers for model training. To address this issue, the concept of FL was introduced. It is a decentralized approach to train models with a server-client architecture. Each client takes a global model from the server and trains the model on its local data without transferring it to the server. This way, no data is transferred as only the model parameters will be shared with the server. The server aggregates the model parameters from its clients and optimizes the global model. This process is repeated until it converges. There are mainly four categories in Federated Learning: Cross-device, Cross-silo, Horizontal, and Vertical. In Horizontal Federated Learning (HFL), we distribute the same model between clients to train on their data with same features having different data samples across clients.

Milavkumar et al. [5] proposed the utilization of a CNN-FL model for Credit Card Transaction Fraud Detection. They achieved an accuracy of 99.86% and applied SMOTE to remove imbalance in their dataset. Stephanie et al. [1] utilized FL to perform fraud detection for data privacy in credit card fraud detections. They achieved 98.4% Binary accuracy and 83.4% Recall. Tanmay et al. [2] proposed various FL approaches for Credit Card Fraud detection in their review. They achieved what's reported as 99.99% accuracy using Random Forest. While Federated Learning (FL) preserves privacy by keeping raw data localized, the exchange of model parameters between clients and the server still introduces potential security risks. To mitigate this, Quantum Encryption (QE), which harnesses the principles of quantum mechanics, provides a theoretically unbreakable method for securing communication.

In this work, we focus on QKD, specifically the BBM92 protocol and MDI-QKD. The BBM92 protocol uses entangled photon pairs, where each photon is distributed to the client (Alice) and server (Bob). Due to quantum entanglement, any attempt to eavesdrop alters the quantum state and is immediately detectable. To further eliminate vulnerabilities in real-world setups, MDI-QKD allows both Alice and Bob to send quantum states to an untrusted third party for measurement. Since no party needs to trust the measurement devices, MDI-QKD provides resilience against detector-based attacks. By integrating these protocols with FL, we enable secure key exchange for encrypting model parameters, ensuring end-to-end confidentiality even in adversarial environments. Building

on advancements in ML, DL, FL, and QKD, we are proposing an integrated approach. In our framework, we built an intrusion-based detection model to detect and mitigate transactional fraud.

1.1 Research Contributions

The key contributions of this work are:

- A novel quantum-enforced federated learning framework, FedQ-Fraud, for secure, private, and real-time financial fraud detection.
- A federated GRU-based deep learning model that achieves 97.47% global accuracy, outperforming LSTM, 1D-CNN and XGBoost by using the MOON algorithm to effectively handle non-IID data.
- An empirical evaluation of QKD protocols establishing that BBM92 provides superior stability, a higher secret key rate (SKR), and a lower quantum bit error rate (QBER) than MDI-QKD for secure model transmission.
- Comprehensive experimental validation via personalization analysis, confusion matrices, and t-SNE visualizations, demonstrating the framework's ability to learn robust fraud patterns across decentralized clients.

2 Paper Organization

The structure of the paper is as follows: Section 2 outlines the overall system architecture along with its mathematical formulation. Section 3 describes the proposed framework in detail, including its core components: Data Acquisition and Preprocessing, Quantum Encryption assisted Federated Learning, and the Quantum Entanglement Layer. Section 4 presents a comprehensive performance analysis, assessing both the effectiveness of the federated learning models and the robustness of the quantum security mechanisms. Finally, Section 5 concludes the paper by highlighting the key contributions and proposing directions for future research.

3 System Model and Problem Formulation

3.1 System Model

The proposed system follows a FL architecture consisting of a central server and multiple distributed clients $\{C_1, C_2, \dots, C_K\}$, each possessing a private, non-IID local dataset. Clients collaboratively train a deep learning model (GRU-based) for fraud detection without sharing raw data, thereby preserving privacy. In each communication round, clients download the global model, update it locally using their data, and send back the model parameters to the server. The server aggregates the updates using weighted averaging to form a new global model using Bonawitz et al.'s Secure Aggregation protocol. To ensure secure transmission of model weights, QKD using the BBM92 protocol is employed to establish secret keys for

encrypting parameters during communication. The global model at round t is updated as:

$$\theta_g^{(t)} = \sum_{k=1}^K \frac{n_k}{n} \theta_k^{(t)}, \quad (1)$$

where $\theta_k^{(t)}$ is the model from client k , n_k is the number of data samples on client k , and $n = \sum_{k=1}^K n_k$ is the total number of samples across all clients. This framework ensures both learning efficiency and communication security.

4 The Proposed Framework

To combat the growing threats in online transactions, the proposed framework is designed to learn and adapt to evolving frauds, thereby detecting them for safer online transactions. We've utilized DL to learn from past experiences, FL for maintaining data privacy, and QEC to ensure a secure communication medium.

4.1 Data Acquisition and Preprocessing Layer

At the start, we acquire data to train our model. Raw data was gathered and preprocessed from a dataset $\mathcal{D} = \{x, y\}^n$ containing 50,000 unique entries of online transactions and 21 features composed of 16,067 (32%) fraudulent and 33,933 (67%) legitimate transactions reflecting a highly imbalanced data. Let the input feature matrix be denoted as $x \in \mathbb{R}^{n \times 21}$ and binary fraud label vector denoted as $y \in \{0, 1\}^n$. Here n is the number of unique entries, x is feature vector of each entry. $y = 0$ shows it is a legitimate transaction, and $y = 1$ shows it is a fraudulent transaction.

$$\mathcal{D} = \{(x, y) | x_i \in \mathbb{R}^{21}, y \in \{0, 1\}\}_{i=1}^n \quad (2)$$

We have used a synthetic financial transaction dataset which resembles real-world transaction patterns while ensuring that it contains no personally identifiable information (PII). The records were injected using rule-based logic such as unusual high transaction amounts, out-of-pattern frequencies which reflects realistic fraud behaviors. the dataset contains the following attributes: Transaction amount, Transaction type, Timestamp, Account balance, location, IP address, Merchant category, Card type, count of past transactions, failed login counts, High risk country, Foreign transaction etc.

- *Data Transformation*: To improve the quality of data, we scale all the data features to a common range to ensure consistency between them.
- *Standard Scaler (SS)*: In this approach, we standardize features by removing the mean value and scaling them to unit variance. It is also known as Z-Score Normalization.

$$x_i^{scaled} = \frac{x_i - \bar{x}}{\sigma} \quad (3)$$

Here, σ is Standard Deviation of the feature, $\bar{x}$ is the mean of the feature. Final Dataset: After imputing missing values and normalization, the processed dataset is represented as:

$$\mathcal{D} = \{x^{final}, y\}^n \quad (4)$$

Where x^{final} is imputed and normalized feature vector for each entry.

4.2 FL Layer

4.2.1 Client Partitioning. In our setup, the dataset is partitioned into 4 clients based on the timestamp (months): Client 1 (Jan–Mar, 12,302 samples), Client 2 (Apr–Jun, 12,513), Client 3 (Jul–Sep, 12,553), and Client 4 (Oct–Dec, 12,634). Each client trained only on its local subset, ensuring Non-IID heterogeneity. Within each client, we applied SMOTE to mitigate fraud imbalance, then stratified into 80% training and 20% test sets.

4.2.2 Baseline Model Selection. To establish a strong baseline for the fraud detection, we first evaluated GRU, LSTM, 1D-CNN and XGBoost to find out the best performing model integrated with the Federated Averaging (FedAvg) algorithm. The client trains the data by applying the Stochastic Gradient Descent (SGD) while the server takes the weighted average of each parameter and updates the model. Let P be the number of participating clients. Let n_c be the number of data samples on client c , and $n = \sum_{c=1}^P n_c$ be the total number of samples across all clients. Let w_c be the model parameters after local training on client c . Then, the global model weights w^{global} after aggregation are:

$$w^{global} = \sum_{c=1}^P \frac{n_c}{n} w_c \quad (5)$$

4.2.3 GRU Neural Network. The GRU model consists of 3 layers enhanced with the federated MOON algorithm for the binary classification between the fraudulent and the legitimate classes. The model architecture has 3 stacked GRU layers of 128 units with the dropout of 20%. In the output layer, we have used CrossEntropyLoss as loss function which internally used softmax by producing logits (raw outputs). For the adaptive optimization of our model, Adam optimizer is used with the learning rate of 0.0005 and regularized weight decay of 0.00001.

$$\mathcal{L} = -z_y + \log \left(\sum_{j=1}^C e^{z_j} \right) \quad (6)$$

where:

- $\mathcal{L}$: Cross-entropy loss
- z_j : Logit for class j
- z_y : Logit corresponding to the true class label y
- C : Total number of classes

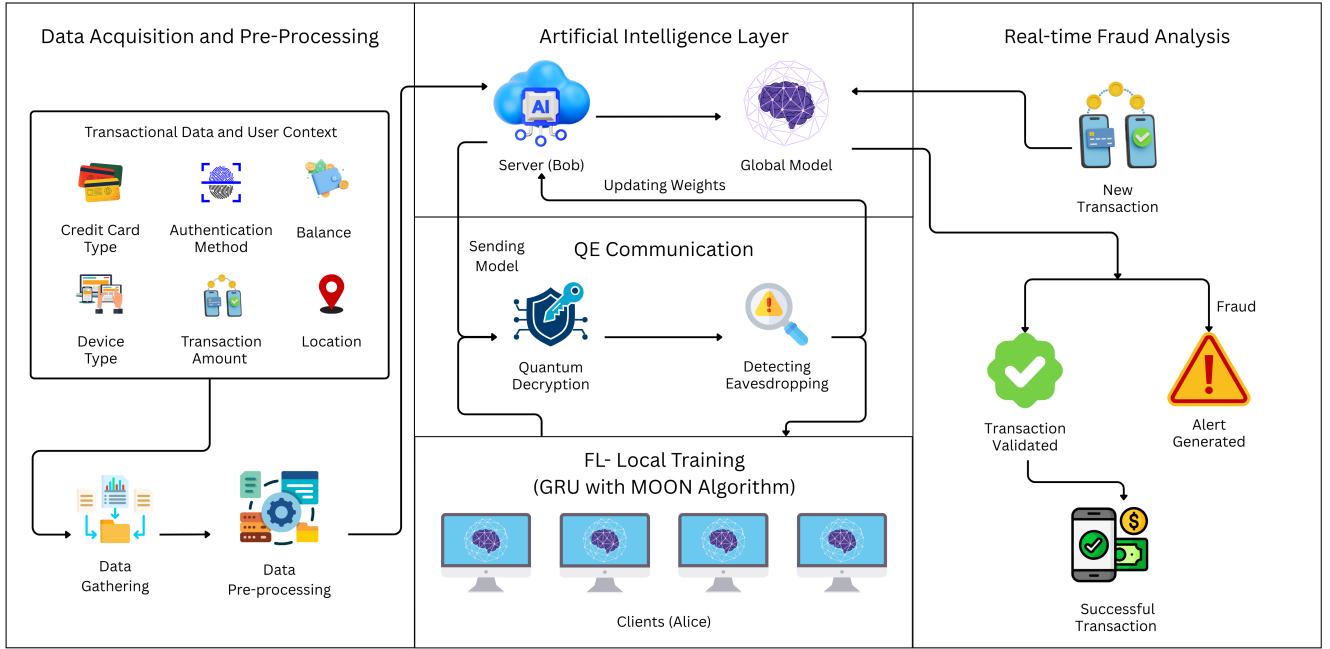


Figure 1: Proposed Framework

4.2.4 Federated Optimization Layer. To address client data heterogeneity, we employ Ditto, a personalized FL algorithm applied over GRU with FedAvg for global aggregation. Each client trains its GRU locally while regularizing updates against the global model. A composite loss with a regularization term penalizes deviations from the global parameters, ensuring balance between personalization and global consistency. This yields a modified local update rule combining both objectives, leading to a new local update rule like:

$$\Phi_i^* = \arg \min_{\Phi_i} \mathcal{L}_{CE}^{(i)}(\Phi_i) + \frac{\lambda}{2} \|\Phi_i - \Phi_g\|^2 \quad (7)$$

- Φ_i : Personalized model parameters of client i .
- Φ_g : Global model parameters aggregated using the FedAvg algorithm.
- Φ_i^* : Optimized personalized model parameters for client i after minimizing the local Ditto objective.
- $\mathcal{L}_{CE}^{(i)}$: Cross-entropy loss computed on client i
- λ : Regularization coefficient
- $\|\Phi_i - \Phi_g\|^2$: Squared ℓ_2 -norm regularization term

Moon (Model Contrastive Federated Learning). It is a personalized federated learning approach that enhances global training by aligning local models with the global model and distinguishing them from previous local states. Moon is used to enhance the FL by integrating contrastive learning to preserve client-specific representations as well as enforces the model consistency across rounds and prevents the client from blindly following

the global model. The contrastive loss gradually pulls the local model towards the global model.

4.3 Quantum Layer for Secure Model Weight Transmission

To ensure secure communication during sensitive stages such as model weight transmission between the server and clients in federated learning, we integrate a quantum-secured communication layer based on QKD. We evaluate two entanglement-based QKD protocols: MDI-QKD and the BBM92 protocol. MDI-QKD offers strong theoretical guarantees by removing trust from measurement devices; both the server (Bob) and client (Alice) send quantum states to an untrusted relay for Bell state measurement. While this approach mitigates detector-based attacks, it introduces practical challenges in synchronization and photon interference. In contrast, the BBM92 protocol, which we ultimately adopt in our final system, demonstrated superior performance in our experiments, particularly in noisy environments. BBM92 relies on the distribution of entangled photon pairs between Alice and Bob. Each party performs measurements using randomly selected bases and publicly announces the basis choices. Bits corresponding to matching bases are retained to form a shared secret key. To detect eavesdropping or channel

Algorithm 1 Federated MOON + GRU with Bonawitz-style Secure Aggregation

Require: Clients $\mathcal{K} = \{1, \dots, K\}$, rounds T , GRU $f(\cdot; \theta)$, contrastive weight μ , quantization scale s , temperature τ

- 1: Server initializes θ_g^0 and broadcasts
- 2: **for** each client $k \in \mathcal{K}$ **do**
- 3: Generate persistent Ed25519 keypair $(\text{sk}_{\text{Ed},k}, \text{pk}_{\text{Ed},k})$
- 4: **end for**
- 5: **for** round $t = 1$ to T **do**
- 6: **Client-side** ($\forall k$):
- 7: Download $\theta_g^{t-1}, \theta_{\text{prev}}^{t-1}$
- 8: Train with MOON loss:

$$\mathcal{L}_k = \mathcal{L}_{\text{CE}} + \mu \mathcal{L}_{\text{contrastive}}$$

$$\mathcal{L}_{\text{contrastive}} = -\log \frac{\exp(\cos(z_k, z_g)/\tau)}{\exp(\cos(z_k, z_g)/\tau) + \exp(\cos(z_k, z_{\text{prev}})/\tau)}$$

- 9: Compute update $\Delta_k^t = \theta_k^t - \theta_g^{t-1}$, flatten δ_k^t , quantize q_k^t
- 10: Generate ephemeral keypair $(\text{esk}_k^t, \text{epk}_k^t)$, sign σ_{epk} , broadcast
- 11: **for** each peer $j \neq k$ **do**
- 12: Derive seed $\text{seed}_{k \leftrightarrow j}^t$, mask $m_{k \leftrightarrow j}^t$
- 13: **end for**
- 14: Form masked payload $\tilde{q}_k^t = q_k^t + \sum_{j \neq k} \text{sgn}(k, j) m_{k \leftrightarrow j}^t$
- 15: Sign σ_k^t and send $(\tilde{q}_k^t, \text{epk}_k^t, \sigma_{\text{epk}}, \sigma_k^t)$
- 16: **Server-side:**
- 17: Verify signatures, collect valid $\mathcal{A}^t$
- 18: Aggregate (SecAgg): $S^t = \sum_{k \in \mathcal{A}^t} \tilde{q}_k^t$
- 19: **if** no dropouts **then**
- 20: $\bar{\Delta}^t = \frac{1}{K} Q_s^{-1}(S^t)$
- 21: **else**
- 22: Reconstruct missing masks C^t , adjust $S^t \leftarrow S^t - C^t$, then $\bar{\Delta}^t = \frac{1}{K} Q_s^{-1}(S^t)$
- 23: **end if**
- 24: Update global $\theta_g^t = \theta_g^{t-1} + \text{Unflat}(\bar{\Delta}^t)$
- 25: **end for**

Ensure: Global θ_g^T and client models θ_k^T

disturbance, we estimate QBER using:

$$Q_{\text{total}} = \frac{1}{r_{\text{sig}} + r_a} \left(q_i r_{\text{sig}} + \frac{1}{2} r_a \right) \quad (8)$$

where r_{sig} is the signal coincidence rate, r_a the accidental coincidence rate, and q_i the intrinsic error probability. If Q_{total} exceeds a predefined threshold θ , the communication session is terminated.

Entangled State Representation: The protocol uses maximally entangled Bell states such as:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad (9)$$

Entanglement Fidelity: The quality of the generated entangled state is quantified using fidelity F , calculated as:

$$F = \langle \Phi^+ | \rho | \Phi^+ \rangle \quad (10)$$

High fidelity ensures secure key generation and robust system performance. Overall, while both MDI-QKD and BBM92 protocols were successfully implemented, the MDI-QKD-based quantum layer demonstrated superior throughput performance, maintaining consistently higher rates across simulation iterations. This enhanced throughput makes MDI-QKD more suitable for high-frequency secure key distribution. However, the BBM92 protocol offered comparatively more stable and predictable behavior, which facilitated smoother integration with our federated learning framework. BBM92's consistency supports the establishment of a tamper-resistant quantum channel for encrypted model weight exchange.

While the integration of QKD ensures secure transport of model weights during federated learning, it does not in itself mitigate the risks of update leakage, where gradients or model updates may reveal sensitive client data. To address this, we complement our QKD-based secure channel with secure aggregation (SA), which ensures that the server only observes aggregated model updates rather than individual client updates. By offering privacy against inference attacks (through SA) and confidentiality in transit (through QKD), this layered defense aligns our system's privacy claims with the mechanisms that are actually in place. Integrating QKD into federated learning introduces both computational and communication overheads. The need for entanglement distribution, photon synchronization, and key sifting processes results in higher latency and lower throughput when compared to classical cryptography. While these factors might limit scalability in large client populations, the security that QKD provides justifies this trade off in domains where data confidentiality is critical, such as healthcare or finance.

5 Results and Discussions

In this section, we present the analysis of our DL-based federated models and different FL algorithms proposed by comparing the evaluation metrics. We compared the evaluation metrics of our FDL models for the selection of the best suitable model among XGBoost, 1D-CNN, GRU, and LSTM. Among these, the GRU model achieved highest global accuracy of 96.32%. Then we assessed various FL algorithms to ensure how the model trains and aggregates the model parameters across the 4 clients. We compared Fedavg, MOON and Ditto algorithms for the evaluation of model aggregation where

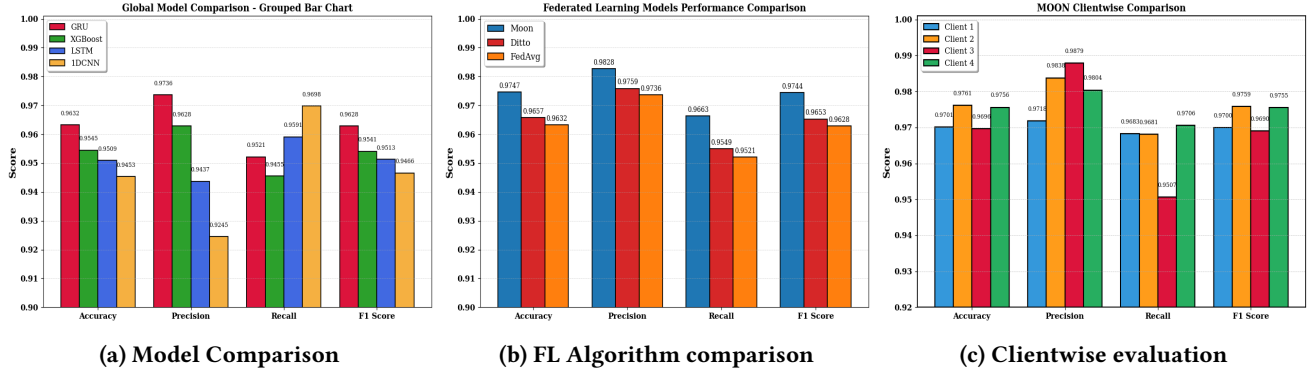


Figure 2: Combined evaluation of model performance and client-wise analysis.

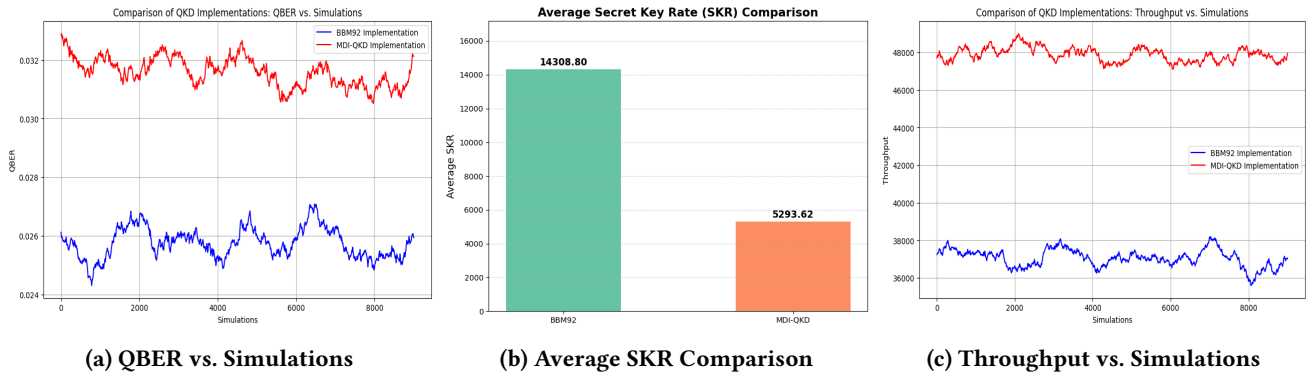


Figure 3: Performance Comparison of BBM92 and MDI-QKD using QBER, SKR, and Throughput

Moon got the highest Global accuracy with 97.47%. So, the GRU-based Moon FL model was evaluated on the basis of accuracy, precision, recall, F1-score, confusion matrix as well as personalization performance evaluation.

5.1 Model Comparison

Figure 2a compares the Global metrics containing accuracy, precision, recall and F1-score of four models (GRU, LSTM, 1DCNN and XGBoost) among which, GRU achieved highest accuracy, precision representing overall strong performance with a balanced fraud detection due to best F1-score. While 1DCNN achieved the highest recall showing the best catching rate of all the frauds.

5.2 FL Algorithm Comparison

Figure 2b compares the Global metrics evaluations of GRU-based FL models. Among them, the Moon-based GRU model achieved the highest scores showing the most balanced and overall strong personalized performance.

5.3 Best Model Performance

Figure 2c compares the per-client evaluation of GRU-based FL model. The plot shows a proper distribution of performance of each client showing the model's consistent performance across the clients.

5.4 Performance Analysis of BBM92 and MDI-QKD Implementations

To evaluate the performance of the two entanglement-based QKD protocols, BBM92 and MDI-QKD, we conducted extensive simulations under identical conditions. The results highlight differences in both QBER and average SKR.

Figure 3a illustrates that BBM92 consistently maintains a lower QBER throughout the simulations, contributing to its higher secret key rate and improved reliability in secure communication. Figure 3b shows that BBM92 yields a significantly higher average SKR (14308.80) compared to MDI-QKD (5293.62). BBM92 demonstrates a superior performance in key generation, making it more efficient for our use case where rapid key generation is critical. Figure 3c demonstrates that MDI-QKD outperforms BBM92 in terms of throughput across simulations.

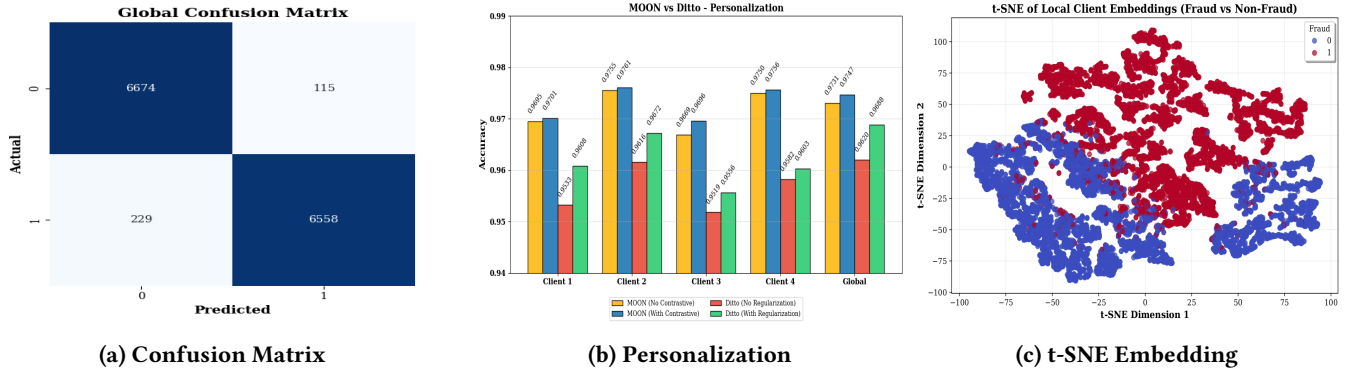


Figure 4: Model Comparison using Confusion Matrix, Personalization, and t-SNE Embedding

This higher throughput indicates more efficient data transmission, making MDI-QKD better suited for applications demanding high-speed secure communication.

5.5 Best Algorithm Performance

Figure 4a shows 97.47% of prediction accuracy of fraud and non-fraud classes across the 4 clients demonstrating an accurate GRU-based model having consistent performance across the clients as well as the server. Figure 4b represents the personalized model comparison between MOON and Ditto showing the effect of personalized training on each client. Figure 4c represents the visualization of the t-SNE which shows the model's ability to generalize across clients by learning discriminative embeddings of fraud and non-fraud classes by forming partially separable clusters. This shows that the GRU model is successfully capturing client-specific behavior as well as global fraud patterns, which enables precise and adaptive detection while preserving data privacy.

6 Conclusion

In this paper, we proposed a secure, privacy-preserving, and high performance fraud detection framework that leverages the strengths of DL, FL, and QKD. The proposed system uses a GRU-based federated learning model enhanced with the MOON algorithm to address the challenges of non-IID data distribution across clients. Comparative evaluations demonstrated that GRU outperformed LSTM, 1D-CNN and XGBoost in classification accuracy and F1-score. Additionally, the MOON-based FL model achieved the highest global accuracy of 97.47%, making it suitable for real world deployment for secure communication. BBM92 and MDI-QKD are two QKD protocols that we implemented and compared for safe model weight transfer in federated learning. While both were effective, BBM92 achieved lower QBER and higher SKR, making it more reliable for communication. By the integration QKD with secure aggregation, our framework

ensures confidentiality in transit, privacy against inference attacks, and reliable fraud detection, contributing to next-generation financial cybersecurity. Future directions include real time fraud detection, adversarial robustness, and extending quantum infrastructure for multi party scalability.

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